Using OLS Regression to Predict Median House Values in Philadelphia

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1 Introduction

In this analysis, we examined the relationship between Philadelphia median house values and several neighborhood characteristics using census block-level data. We measured the relationship by regressing median house value on the following variables: - Proportion of residents in Block Group with at least a bachelor's(PCTBACHMORE) - Proportion of housing units that are vacant (PCTVACANT) - Percent of housing units that are detached single family houses (PCTSINGLES) - Number of households with incomes below 100% poverty level (NBELPOV100)

The regression output tells us that the there is a significant relationship between our dependent variable (Median house value) and each predictor (PCTVACANT, PCTSINGLES, PCTBACHMORE, and LNNBELPOV100). Holding all other variables constant:

- A 1 percentage point increase in PCTVACANT is associated with a 1.9% decrease in median house value (p-value < 2e-16)
- A 1 percentage point increase PCTSINGLES is associated with a .29% increase in median house value (p-value = 2.42e-05)
- A 1 percentage point increase in PCTBACHMORE is associated with a 2% increase in median house value (p-value < 2e-16)
- A 1 percent increase in LNNBELPOV100 is associated with a .79% decrease in median house value (p-value < 2e-16)

Each of these variables has a p-value well below .05, meaning that we can confidently say that if there is actually no relationship in the population between the dependent variable and each of the independent variables, then the probability of getting sample beta coefficients this extreme is less than the associated p-value.

These low probabilities illustrate that we can safely reject:

$H_0 : \beta_1 = 0$ for $H_1 : \beta_1 \neq 0$, $H_0 : \beta_2 = 0$ for $H_a : \beta_2 \neq 0$, $H_0 : \beta_3 = 0$ for $H_a : \beta_3 \neq 0$, and $H_0 : \beta_4 = 0$ for $H_a : \beta_4 \neq 0$.

About 66% of the variance in the dependent variable is explained by the model (R^2 and Adjusted R^2 are .6623 and .6615, respectively). The low p-value associated with the F-ratio illustrates that we can reject the null hypothesis that all coefficients in the model are 0.

Peer-reviewed studies suggest a positive relationship between median house value and median household income, and neighborhoods with clustered college graduates (Gilraine et al.; Alexander and Whitney; Nelson et al.) . We want to examine a broader set of variables and their relationship to median house value.

Historically, single-family homes are clustered in areas with higher median house value. Starting in the 1930s, racist redlining and zoning practices became commonplace across the country, resulting in the further segregation of neighborhoods, the denial of Black and brown families a path to generational wealth accumulation, and the mass displacement of historically Black and brown communities. White and wealthy neighborhoods, typically with a large concentration of single-family homes, higher rates of education, and higher median household income, were protected from harmful redevelopment such as the bisection of communities through highway development

and eminent domain of homes and businesses. Meanwhile, Black and brown communities were subject to significant displacement and deliberately denied access to homeownership and property value appreciation.

Although race is not a variable in our model, Philadelphia's history of redlining and Urban Renewal contributed to the economically stratified neighborhood patterns we see in this model. The history of redlining and displacement does not disaggregate race from class; in fact, the impact of these policies were racist in their institutional denial of upward mobility in marginalized communities and classed in their racialization. Policies such as redlining and urban renewal were framed as economic revitalization initiatives, but in practice, they were disinvestment initiatives in Black and brown neighborhoods, making poor neighborhoods poorer and wealthy neighborhoods wealthier. The predictors we evaluate are likely related to our dependent variable because they stem from the same historical processes that shaped the residential construction of Philadelphia and much of our country.

2 Methods

2.1 Data Cleaning

This analysis uses 2000 Census data for Philadelphia at the block group level. The original dataset contained 1,816 block groups. Four categories of observations were removed before the data were released for analysis: block groups with a population below 40, block groups with no housing units, block groups with a median house value below $10,000, and one North Philadelphia block group reporting a median house value above $800,000 alongside a median household income below $8,000. In each case the records would have produced unstable or implausible estimates, whether because a median drawn from very few households is easily distorted, because no housing market exists to measure, or because the reported values are inconsistent with the rest of the city.

The cleaned dataset contains 1,720 block groups. No further outlier screening was carried out here, and all analyses that follow are based on these 1,720 observations. The variables used are:

- **POLY_ID**: Census Block Group ID
- **MEDHVAL**: Median value of all owner occupied housing units
- **PCTBACHMOR**: Proportion of residents in Block Group with at least a bachelor's degree
- **PCTVACANT**: Proportion of housing units that are vacant
- **PCTSINGLES**: Percent of housing units that are detached single family houses
- **NBELPOV100**: Number of households with incomes below 100% poverty level (i.e., number of households living in poverty)
- **MEDHHINC**: Median household income

2.2 Exploratory Data Analysis

2.2.1 Summary Statistics and Distributions

Before fitting any model, we examine the summary statistics and distributions of the dependent variable and each predictor. For every variable we calculate the mean and the standard deviation, and we plot a histogram to assess the shape of its distribution. Where a distribution is strongly skewed, we apply a natural log transformation and re-examine the histogram to determine whether the transformed variable is better behaved than the original. Variables containing zero values are transformed as the natural log of one plus the variable, since the log of zero is undefined. We also plot the dependent variable against each predictor to assess whether the relationships appear linear, and we map the dependent variable and the predictors across the city to look for spatial patterns.

2.2.2 Correlations Between Predictors

As part of this exploratory stage we also examine the correlations between the predictors. A correlation measures the strength and direction of the linear association between two variables, that is, whether they tend to move together, move in opposite directions, or show no consistent linear pattern at all. The sample correlation coefficient r can be written as the average product of the standardized deviations of the two variables,

$$r = \frac{1}{n-1} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right) \tag{1}$$

$r =$ sample correlation coefficient
$n =$ number of observations, here the 1,720 block groups
$x_i,\ y_i =$ values of the two variables for observation i
$\bar{x},\ \bar{y} =$ sample means of x and y
$s_x,\ s_y =$ sample standard deviations of x and y

Because each deviation from the mean is divided by the standard deviation of its own variable, r is unitless and can be compared across variables measured on entirely different scales.

The correlation coefficient ranges from -1 to $+1$. A value of $+1$ indicates a perfect positive linear relationship, in which the two variables increase together in exact proportion, and a value of -1 indicates a perfect negative linear relationship, in which one increases exactly as the other decreases. A correlation of 0 means there is no linear association between the two variables, though it does not rule out a relationship of some other, non-linear form.

We examine these correlations because strongly inter-correlated predictors create multicollinearity, which makes it difficult for the regression to isolate the individual contribution of each predictor and leaves the coefficient estimates unstable. A correlation greater than $|0.8|$ between a pair of predictors would indicate severe multicollinearity and call for dropping one of them. The

dependent variable is deliberately excluded from this check, since a strong correlation between a predictor and the dependent variable is desirable rather than problematic.

2.3 Multiple Regression Analysis

2.3.1 The Method

Multiple regression is used to model the relationship between a single continuous dependent variable and a set of predictors. It estimates a linear equation that describes how the dependent variable changes as each predictor changes, and it does so for all predictors simultaneously rather than one at a time. The result is a coefficient for each predictor that represents its association with the dependent variable while the remaining predictors are held constant, which allows the individual contribution of each predictor to be isolated. The fitted equation can then be used both to interpret those relationships and to generate predicted values for observations in the data.

2.3.2 The Model

For this analysis, the dependent variable is the natural log of median house value, and the model is

$$\begin{aligned} \text{LNMEDHVAL} = \beta_0 + \beta_1\,\text{PCTVACANT} + \beta_2\,\text{PCTSINGLES} \\ + \beta_3\,\text{PCTBACHMOR} + \beta_4\,\text{LNNBELPOV100} + \varepsilon \end{aligned} \tag{2}$$

$\text{LNMEDHVAL} =$ natural log of median house value
$\text{PCTVACANT} =$ percent of housing units that are vacant
$\text{PCTSINGLES} =$ percent of housing units that are detached single family homes
$\text{PCTBACHMOR} =$ percent of residents with at least a bachelor's degree
$\text{LNNBELPOV100} =$ natural log of one plus the number of households in poverty
$\beta_0 =$ the intercept, the expected value of LNMEDHVAL when all predictors equal zero
$\beta_1 \ldots \beta_4 =$ the slope coefficients, each giving the expected change in LNMEDHVAL for a one unit increase in its predictor, holding the others constant
$\varepsilon =$ the error term, the portion of LNMEDHVAL not explained by the predictors

Both the dependent variable and the poverty predictor are log transformed. The remaining three predictors enter the model untransformed, so their coefficients describe the effect of a one percentage point increase in each.

2.3.3 Assumptions

The model rests on four assumptions.

Linearity. The relationship between each predictor and the dependent variable is assumed to be linear. If the true relationship curves, a straight line will misstate the effect across parts of the range.

Independence of observations. Each observation is assumed to be independent of every other. With spatial data this assumption is often questionable, since nearby block groups may resemble one another, a condition known as spatial autocorrelation.

Normality of residuals. The errors are assumed to be normally distributed with a mean of zero. Note that this applies to the residuals rather than to the raw variables, which need not be normal themselves.

Homoscedasticity. The variance of the errors is assumed to be constant across all levels of the predicted values. If instead the errors widen or narrow systematically, a condition called heteroscedasticity, the standard errors and the significance tests built on them may be unreliable.

2.4 Additional Analyses

2.4.1 Stepwise Regression

Stepwise regression automates the choice of which predictors to retain. Starting from a specified model, the algorithm adds or removes one predictor at a time and scores each candidate model using the Akaike Information Criterion, a measure that balances fit against the number of predictors used. Fitting better lowers the score and each additional predictor raises it, so a predictor earns its place only if the improvement outweighs the added complexity. Lower is better, and the value means something only in comparison with other models. The procedure makes whichever move lowers the score most and stops when no further change improves it. We run it in both directions, so predictors may be either added or dropped.

Its limitations is as it searches many models against the same data and reports the winner, it is prone to overfitting, and the p-values for the selected model are optimistic since they ignore the search that produced it. It finds a model that cannot be improved by a single step, not necessarily the best one. It also ignores theory about which predictors should matter, so it is better treated as one piece of evidence than as a decision rule.

2.4.2 K-Fold Cross-Validation

Cross-validation assesses how well a model predicts observations it was not fitted on. This matters because a model is estimated using the full dataset and will necessarily fit that data reasonably well, which makes in-sample measures such as R^2 an optimistic guide to performance on new observations.

We use k-fold cross-validation with $k = 5$. The 1,720 block groups are divided at random into five roughly equal groups, or folds. The model is then fitted five times. On each repetition, four folds are used to estimate the coefficients and the remaining fold is held out, with the fitted model used

to generate predictions for the held-out observations. Each fold serves as the held-out set exactly once, so every block group receives a prediction from a model that was estimated without it.

Model performance is summarized by the root mean square error, calculated from these cross-validated predictions:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2} \tag{3}$$

where

$$\begin{aligned} n &= \text{the number of observations, here the 1,720 block groups} \\ y_i &= \text{observed value of the dependent variable for observation } i \\ \hat{y}_i &= \text{cross-validated predicted value for observation } i \end{aligned}$$

Each prediction error is squared so that positive and negative errors do not offset one another and so that larger errors carry disproportionate weight. These squared errors are averaged, and the square root returns the result to the units of the dependent variable, here log median house value. A lower RMSE indicates predictions closer to the observed values, so the measure is interpreted as the typical size of the model's prediction error.

An individual RMSE value is not meaningful in isolation, since its scale depends on the units of the dependent variable. It is used instead to compare models fitted to the same dependent variable. We apply this procedure to the four predictor model specified above and to a reduced model containing only PCTVACANT and MEDHHINC as predictors, and compare the two resulting RMSE values.

2.5 Software

All analysis was carried out in R. Summary statistics, histograms, scatter plots, correlations, and the regression models were produced using base R functions, including `lm` for the regression and `cor` for the correlation matrix. The `MASS` package was used for the stepwise regression, the `DAAG` package for the k-fold cross-validation, and the `sf` and `tmap` packages for reading the shapefile and producing the choropleth maps. The report itself was written in R Markdown, so the code, output, and text are contained in a single document.

3 Results

3.1 Exploratory Results

Table 1: Summary statistics for the dependent variable and predictors

Variable	Mean	SD
Dependent Variable		
MEDHVAL	66,287.73	60,006.08
Predictors		
PCTBACHMOR	16.08	17.77
MEDHHINC	31,541.76	16,298.43
PCTVACANT	11.29	9.63
PCTSINGLES	9.23	13.25
NBELPOV100	189.77	164.32

The dependent variable in this statistical analysis is MEDHVAL, representing the median value of owner-occupied housing units. The independent variables (predictors) are PCTBACHMOR, the proportion of residents in a block group with at least a bachelor's degree; PCTVACANT, the proportion of housing units that are vacant; PCTSINGLES, the proportion of housing units classified as detached single-family homes; NBELPOV100, the number of households with incomes below 100% of the federal poverty level; and MEDHHINC, median household income.

The descriptive statistics indicate substantial variability across Philadelphia block groups in housing values, educational attainment, housing characteristics, poverty, and household income. MEDHVAL has a mean of $66,287.73 and a standard deviation of $60,006.08, indicating considerable variation in median owner-occupied housing values across census tracts. The relatively large standard deviation compared with the mean suggests substantial geographic disparities in housing values.

PCTBACHMOR has a mean of 16.06 and a standard deviation of 17.80. This indicates that, on average, approximately 16% of residents in a block group have at least a bachelor's degree, with considerable variation in educational attainment across the study area. PCTVACANT has a mean of 11.29 and a standard deviation of 9.63, suggesting that approximately 11% of housing units are vacant in the average census tract, although vacancy rates vary by 9% across census tract. Similarly, PCTSINGLES has a mean of 9.23 and a standard deviation of 13.25, indicating that detached single-family homes account for approximately 9% of housing units in the average census tract, with substantial variation in their prevalence.

NBELPOV100 has a mean of 189.77 households and a standard deviation of 164.31 households. Thus, the average census tract contains approximately 190 households with incomes below the federal poverty level, although the number varies considerably across block groups. Finally, MEDHHINC has a mean of $31,541.76 and a standard deviation of $16,298.43, indicating notable differences in median household income across the study area.

Overall, the descriptive statistics suggest considerable heterogeneity among Philadelphia block groups across all six variables. These differences provide important context for examining the relationships between housing values and the selected socioeconomic and housing-related predictors.

Preliminary examination of the variable distributions indicated that several variables deviated

from a normal distribution. To address this issue, we applied logarithmic transformations to each variable in an attempt to reduce skew and improve the distributions' symmetry. The transformations improved the distributions of MEDHVAL and NBELPOV100, which appeared more normally distributed following transformation. However, the remaining variables continued to exhibit pronounced spikes at zero, suggesting that the logarithmic transformation did not adequately normalize their distributions. The following histograms illustrate the distributions of the transformed variables and demonstrate the extent to which the transformations affected their shapes.

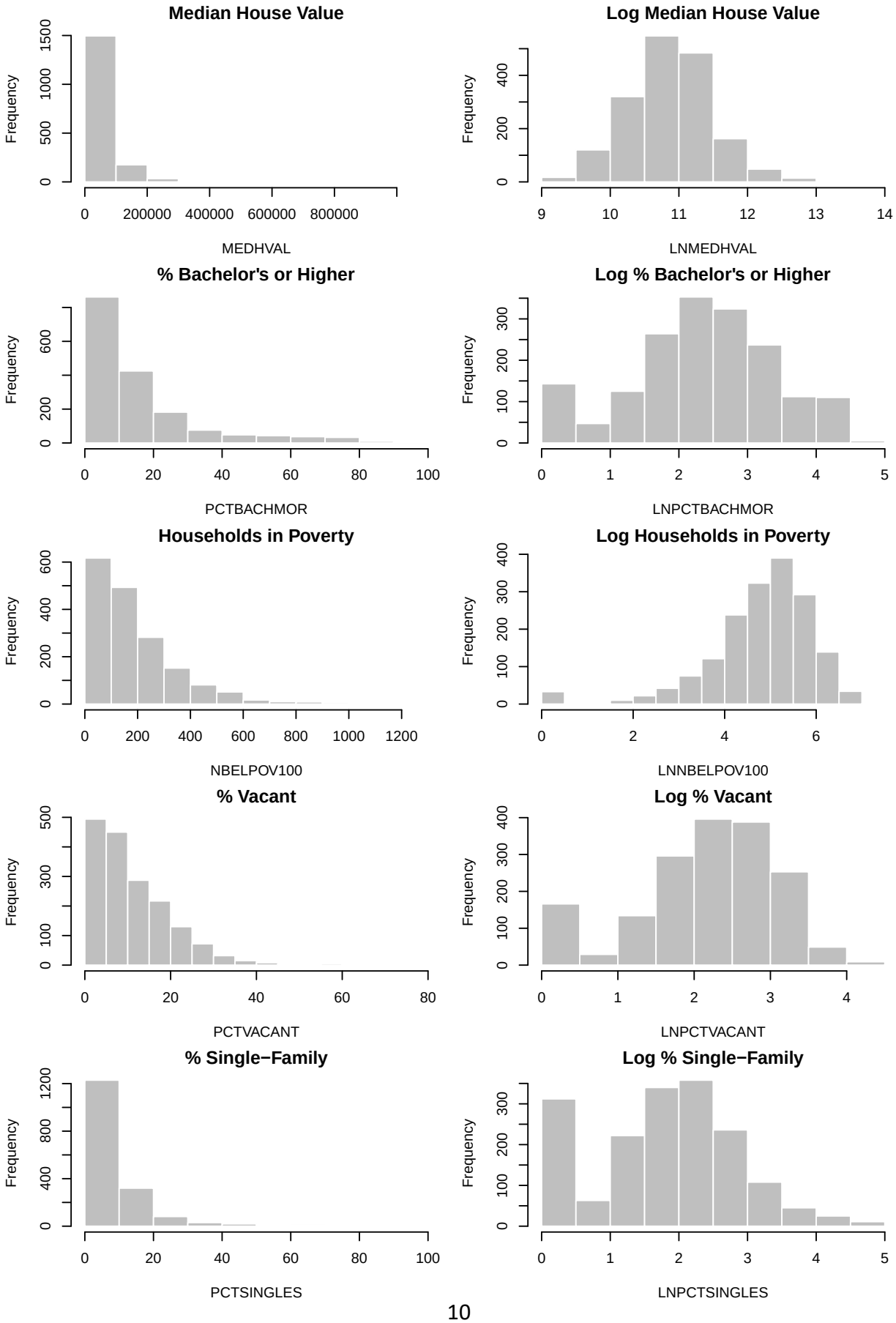


Figure 1: Distributions of the dependent variable and predictors: original values (left) and log-transformed values (right)

Ultimately, we decided to use the original PCBACHMORE, PCTVACANT, and PCTSINGLES variables and the transformed LNMEDHVAL and LNNBELPOV100 variables in the regression. The other regression assumptions will be examined in a separate section below.

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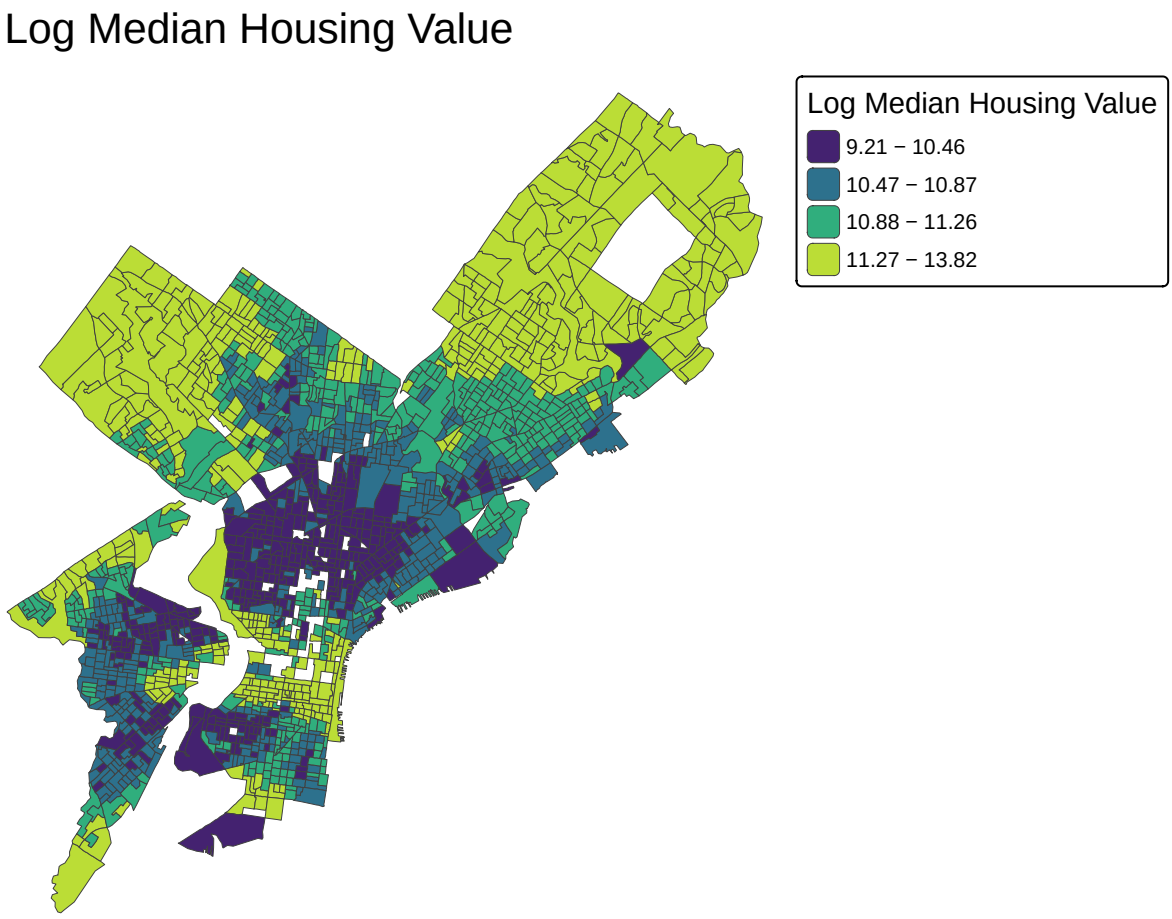


Figure 2: Log median house value by block group

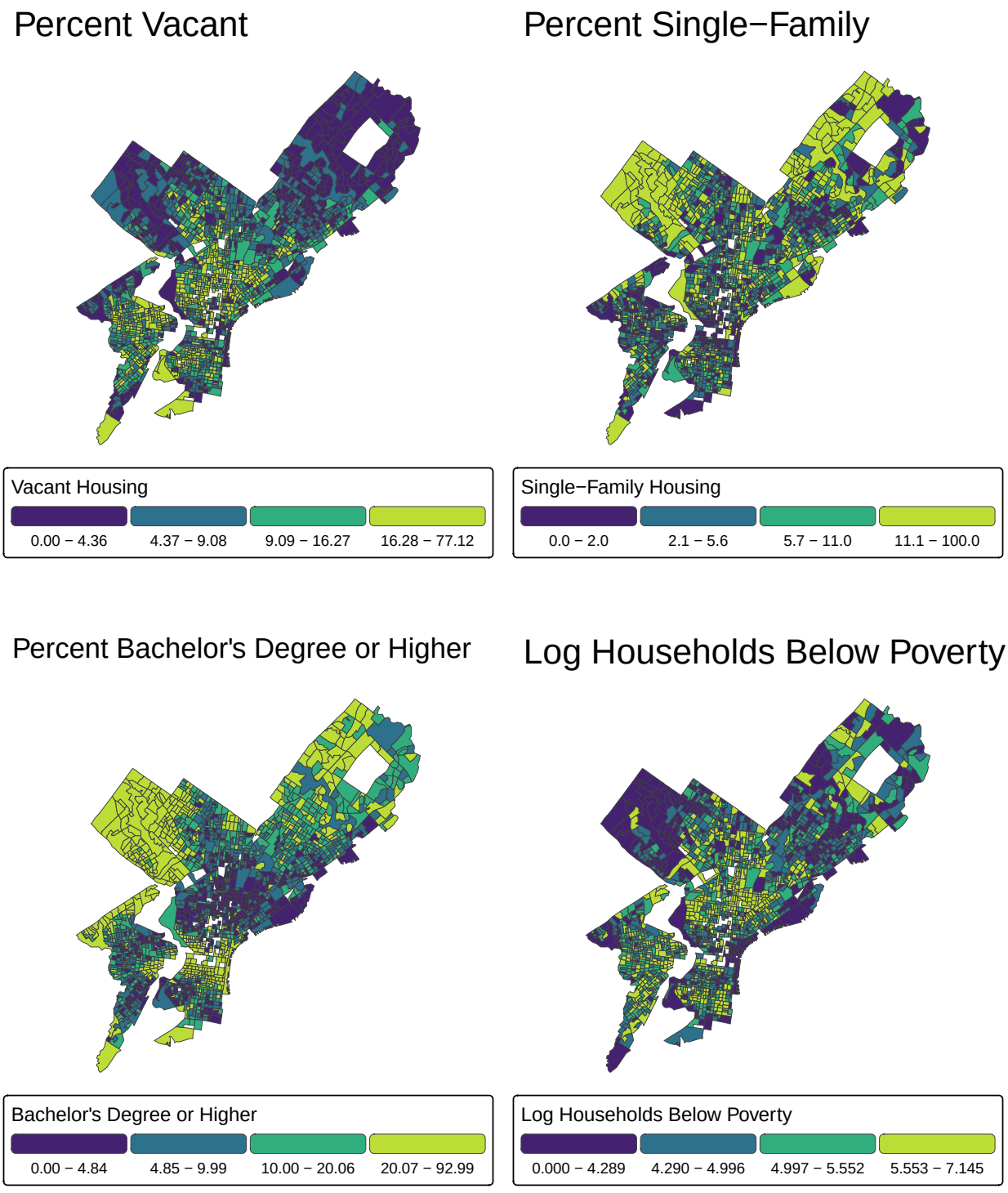


Figure 3: Predictors by block group: percent vacant, percent single-family, percent with a bachelor's degree or higher, and log households below poverty

The choropleth maps that visualize the percent vacant households and the households below

poverty look the most similar…

The choropleth maps that visualize the percent vacant households and the households below poverty look the most similar. The choropleth maps for percent bachelor's degree or higher and percent single-family homes have vague spatial similarities but are not as similar as the former are to each other. This suggests that percent vacant and households below poverty may be inter-correlated and percent single-family and percent bachelor's degree or higher may be inter-correlated. That being said, the maps are similar but not enough that we would expect severe multicollinearity. Based on the maps, we would expect that percent bachelors degree or higher would be the most strongly associated with the dependent variable, then percent single family home.

Table 2: Pearson correlations between the predictors

	PCTVACANT	PCTSINGLES	PCTBACHMOR	LNNBELPOV100
PCTVACANT	1.000	-0.151	-0.298	0.250
PCTSINGLES	-0.151	1.000	0.198	-0.291
PCTBACHMOR	-0.298	0.198	1.000	-0.320
LNNBELPOV100	0.250	-0.291	-0.320	1.000

The correlation matrix does not show severe multicollinearity. The greatest value is -0.3197668. The correlation matrix supports the conclusion made based on the visual comparisons of predictor maps. The correlation matrix suggests that percent vacant and households below poverty have the highest correlation and percent single family households and percent with a bachelor's degree or higher have the second highest correlation. Percent vacant and percent with a bachelors or higher have low correlation, and percent bachelor's or higher and households below poverty have the lowest correlation.

3.2 Regression Results

Table 3: OLS regression results ($R^2 = 0.6623$, adjusted $R^2 = 0.6615$, $F = 840.9$, $n = 1720$)

Term	Estimate	Std. Error	t value	p-value
(Intercept)	11.1138	0.0465	238.8364	< 0.001
PCTVACANT	-0.0192	0.0010	-19.5903	< 0.001
PCTSINGLES	0.0030	0.0007	4.2335	< 0.001
PCTBACHMOR	0.0209	0.0005	38.4939	< 0.001
LNNBELPOV100	-0.0789	0.0085	-9.3303	< 0.001

The regression model is statistically significant, with an F-statistic of 840.9 and a p-value less than 2.2×10^{-16}. The model hads an R^2 of 0.6623 and an adjusted R^2 of 0.6615, indicating that approximately 66.23% of the variation in log median household value was explained by the four

predictors. The adjusted R^2 is very close to the R^2, suggesting that the model's explanatory power remained similar after accounting for the number of predictors. All four independent variables are statistically significant at the 0.001 level, holding the other variables constant. The percentage of vacant housing has a negative coefficient of -0.01916 ($p < 0.001$). This indicates that a one-percentage-point increase in vacant housing was associated with approximately a 1.90% decrease in median household value. The percentage of single-person households has a positive coefficient of 0.00298 ($p < 0.001$), indicating that a one-percentage-point increase in single-person households is associated with approximately a 0.30% increase in median household value. The percentage of residents with a bachelor's degree or higher has a positive coefficient of 0.02091 ($p < 0.001$). This suggests that a one-percentage-point increase in residents with a bachelor's degree or higher was associated with approximately a 2.11% increase in median household value. The natural logarithm of the number of residents living below the poverty threshold has a negative coefficient of -0.07891 ($p < 0.001$). Because both the dependent variable and this predictor were log-transformed, the coefficient can be interpreted as an elasticity: a 1% increase in the neighborhood poverty measure is associated with approximately a 0.079% decrease in median household value. The results indicate that residential vacancy and neighborhood poverty were negatively associated with median household value, while the percentage of single-person households and the percentage of residents with a bachelor's degree or higher were positively associated with median household value. These relationships are statistically significant within the fitted model, although they do not establish causation.

3.3 Regression Assumption Checks

In this section, we will discuss testing model assumptions. We have already looked at the variable distributions earlier.

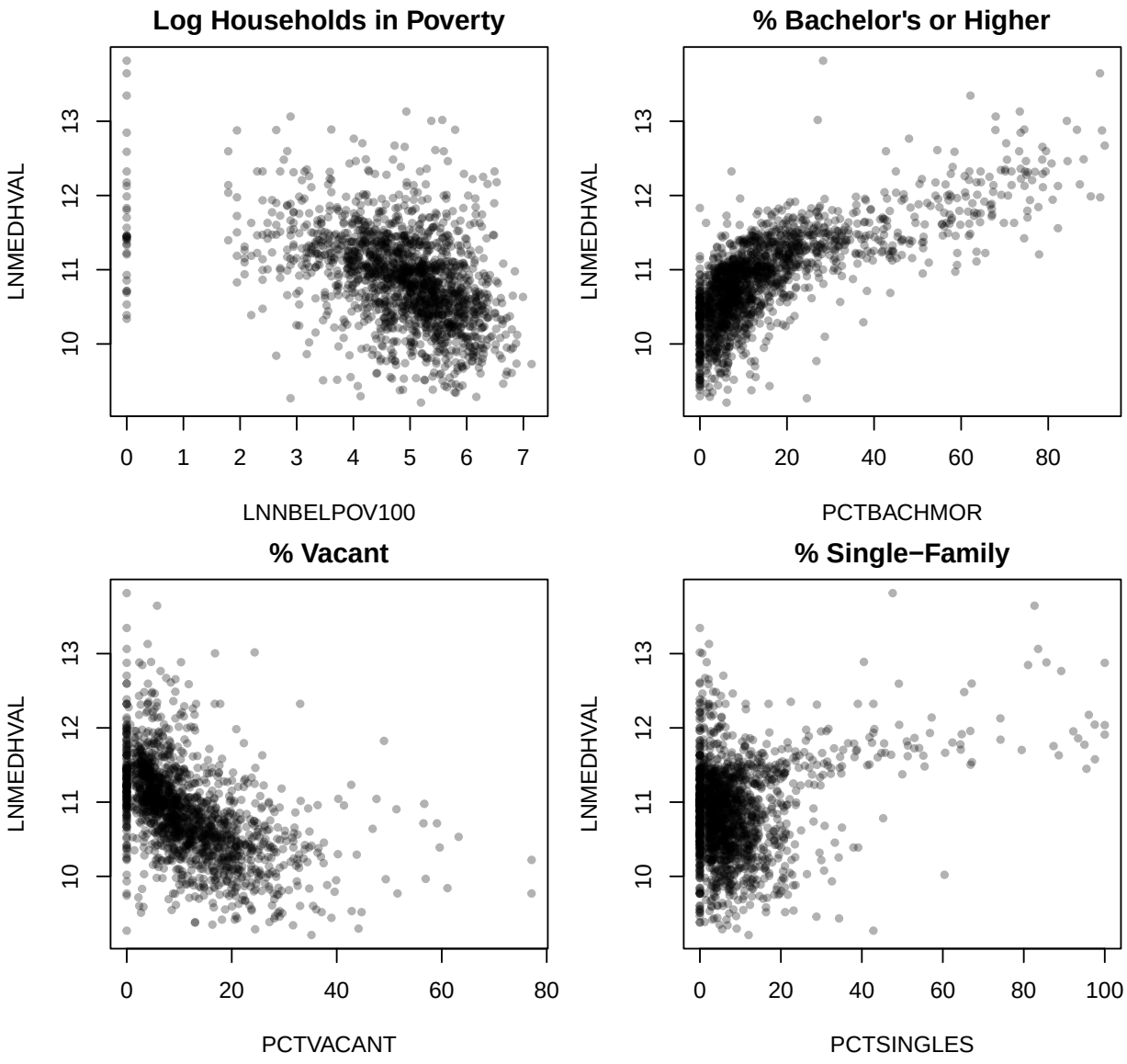


Figure 4: Log median house value plotted against each predictor

The relationships presented by the scatter plots of the dependent variable and each of the predictors does not appear to be linear, as assumed by the regression model. This suggests that the model may not accurately or fully understand the relationships in the data thus reducing its accuracy. Further examination of the residuals will need to be done in order to come to a conclusion.

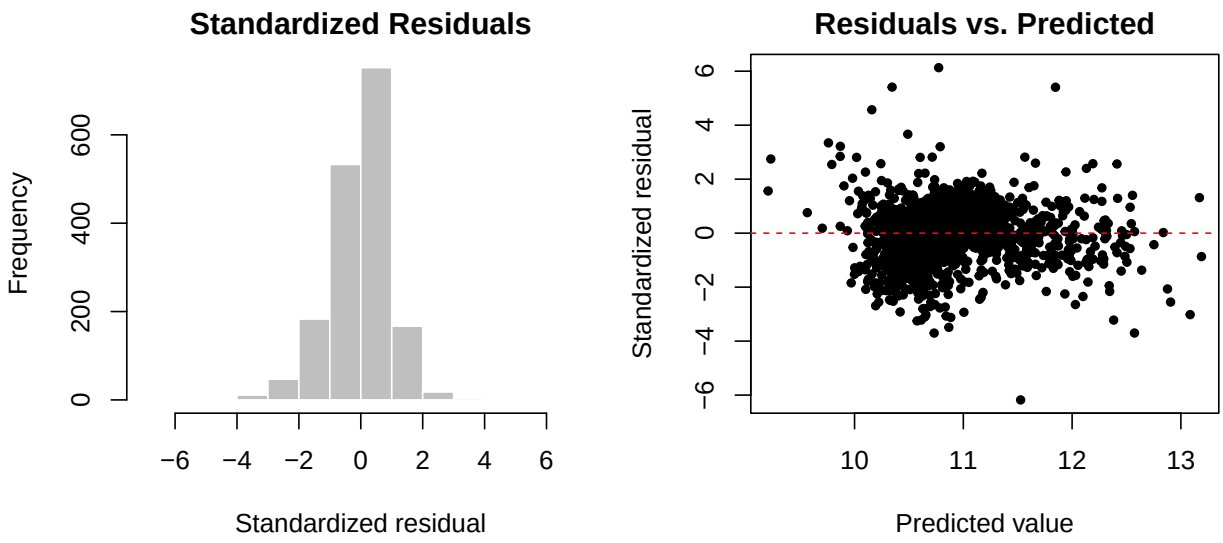


Figure 5: Histogram of standardized residuals (left) and standardized residuals versus predicted values (right)

The histogram of standardized residuals (left) appears to be normally distributed. It is important to check if standardized residuals are normally distributed because it supports the reliability of the statistical test. A non-normal distribution may suggest that there are strong outliers that should be addressed further.

The plot on the right shows the standardized residuals vs. predicted value. Standardized residuals are the differences between the actual data values and the values predicted by the model. It is crucial to plot these against each other in order to determine if there are any unaccounted for predictors. The data is like plot appears as an organic-cloud-like form which indicates that there are likely not predictors unaccounted for, this is pattern is known as homoscedasticity. A heteroscedastic plot would show a pattern in the data, for example the data may fan out toward one side. This would indicate that there are more predictors that need to be included in the regression model. There appears to be some minor outliers at -6 and 6 which indicates that there are a few tracts where the model greatly overpredicts or underpredicts the median household value. This could be caused by unsual census tracts or lapses in the data collection.

There appears to be some spatial autcorrelation in our variables. Across four of the maps, the dependent variable and three of the four predictors, similar results seem to cluster together in nearby locations. The only map that appears to have no or limited spatial autocorrelation is the map of percentage of single family households. This is important to note because spatial inter-dependence may violate the assumptions of the regression module, that all census tracts operate independently. Spatial autocorrelation tests would need to be done in order to draw further conclusion. We will examine the spatial autocorrelation of the variables in residuals in the next assignment.

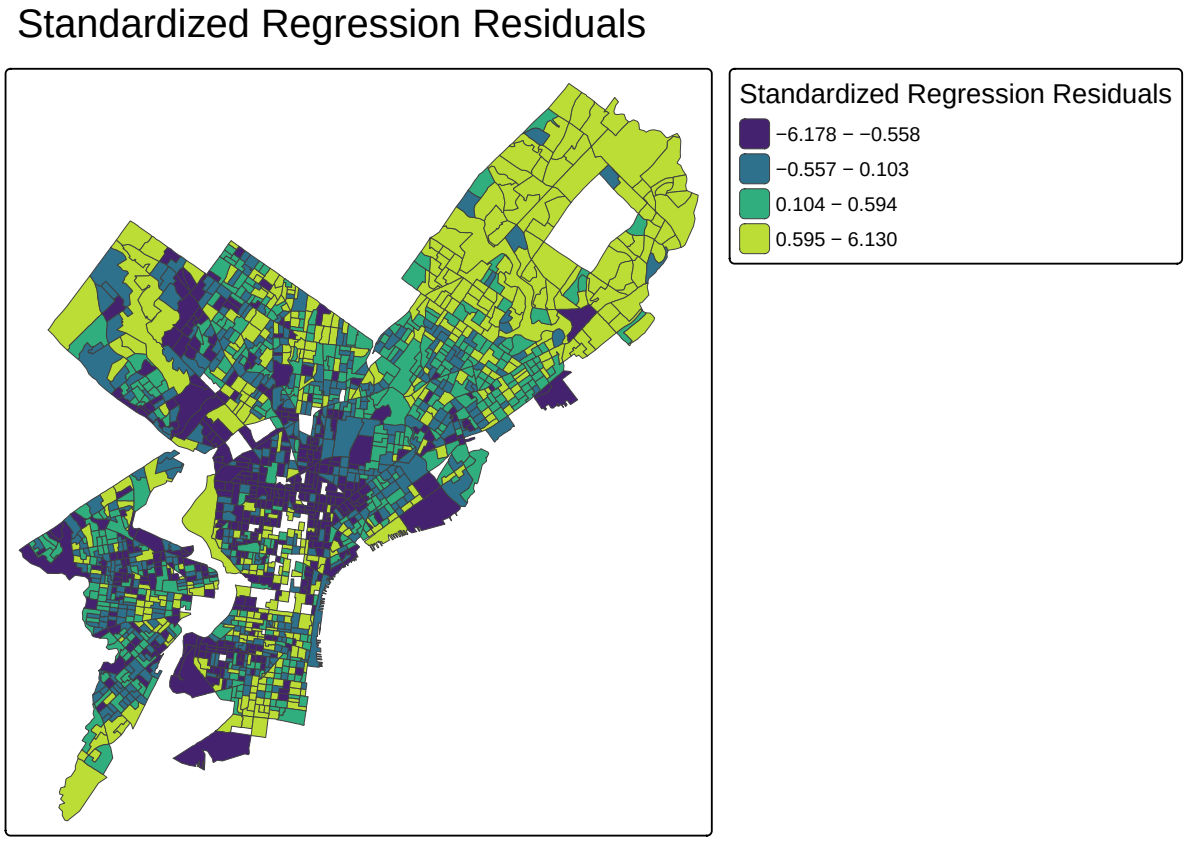


Figure 6: Standardized regression residuals by block group

It seems that there is some spatial autocorrelation in the standardized regression residuals map. Areas towards center city tend to have larger negative values whereas areas towards the borders of the city boundary have larger positive values. Specifically in the northeast, there are many neighboring census tracts that fall within the same value bin. Throughout the rest of the city, the values vary slightly more from tract to tract, though largely follow the pattern described earlier.

3.4 Additional Models

The stepwise regression procedure using the Akaike Information Criterion (AIC) selected the initial model as the final model. All four predictors—PCTVACANT, PCTSINGLES, PCTBACHMOR, and LNNBELPOV100—were retained because removing any individual predictor increased the AIC. The full model had an AIC of 1435.08, the lowest among the models evaluated. Therefore, the procedure did not identify an improvement by removing any of the predictors.

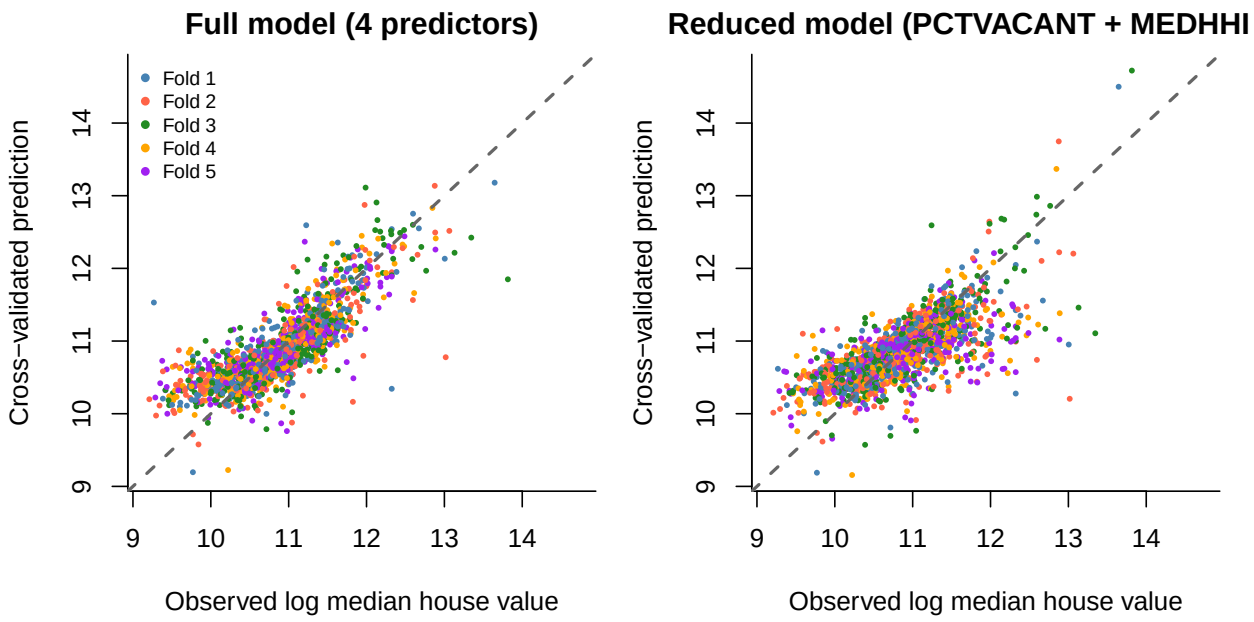


Figure 7: Observed versus cross-validated predicted values from 5-fold cross-validation for the full model (left) and the reduced model (right). The dashed line marks perfect prediction.

The cross-validation RMSE for the original model (0.3664) is lower than the cross-validation RMSE for the model with only two predictors (0.4427), which indicates that the original model is more accurate.

4 Discussion and Limitations

The final model includes all 4 predictors because the stepwise regression based on AIC retained all four predictors. Removing any of them increased the AIC. The RMSE with all four predictors was .37, the RMSE with only two predictors was higher at .44. This tells us that our model with all four variables has a smaller prediction error, which means it predicted new data more accurately. The R^2 and adjusted R^2 were .6623 and .6615, respectively. These are almost identical because our sample is large compared to our number of predictors, so the penalty for many predictors is very small. This indicates a reasonably good model because the four predictors together account for about 66% of the variance in our dependent variable. The F-statistic is 840.86 (p<.001), making us confident that we can reject the null hypothesis.

Most notably, because block groups are spatially clustered, the OLS residuals are most likely spatially autocorrelated. Because an assumption of OLS is independence of observations, spatial autocorrelation–typical with neighborhood data–poses somewhat of a problem for us. It may be advantageous to run a spatial lag or spatial error model to build the neighborhood relationships into the model itself and help us determine if the p-values we calculated from our OLS model are accurate or if they are inflated. We could test for spatial autocorrelation in the residuals using Moran's I and based on the outcome, leverage spatial lag or spatial error model to account for the dependence.

Using raw numbers for households living in poverty may not provide the most accurate representation of rates of poverty in Philadelphia block groups. Two very different-sized block groups can have the same raw number of households living in poverty. By using this data, we don't see the relationship between block group size and rate of households in poverty, we would only see that two block groups have the same raw number of households in poverty. On the other hand, a small block group could have very few households living in poverty, but a high rate, and a large block group could have many households living in poverty but may have a low rate. A percentage of households living in poverty per block group would more accurately capture how concentrated poverty is in each block group.

It would not make sense to run a Ridge or LASSO regression here because these methods are intended to be used on models with severe multicollinearity or a proportionally high number of variables compared to the number of observations. Both methods penalize large coefficients to reduce variance and overfitting. The Ridge regression shrinks and retains every variable, with the goal of stabilizing estimates when there is significant correlation between predictors. The LASSO regression may shrink some coefficients to zero, completely dropping them from the model if they are redundant or ineffective. Because we did not see severe multicollinearity (the most severe r value was -.32) and we have 1,720 observations with only four independent variables, we determined that using a Ridge or Lasso method would not be effective in making our model more predictive and may make our coefficients harder to interpret.

We were surprised that the variables were less tangled together than we initially thought. No pairs exceeded an r-value of |.32|, even though they stem from the historical process described in our introduction. Single-family homes, specifically, has a relatively small effect. In looking at our maps, we noticed that some of the most expensive areas are in dense neighborhoods like Rittenhouse and Society Hill, which are largely comprised of row homes, which is unique to Philadlephia and other mid Atlantic cities like Baltimore. Because our predictors do not account for a history of redlining and racist zoning practices we saw in other data, its impact may be in the margins of our model. Could it be explained by other factors? Our residual map may more potently tell the story of redlining. It seems to show that our model overestimates historically Black and brown neighborhoods and underestimates historically whiter neighborhoods. Since we did not explicitly include race in our model, this is still just an estimation of what other models may or may not tell us. We are interested in comparing this model to a spatial lag or a spatial error model to test this theory directly.

5 Sources

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